

Pre-Feasibility Study for Automatic Parking in Irbid



2017

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1. Executive Summary

The city of Irbid is an attractive area for university students from inside and outside Jordan due to the large number of universities. It is also characterized by the presence of many archaeological and tourist areas, making it a city suffering from traffic congestion due to the spread of institutions, companies and commercial centers in addition to the increase in the number of vehicles in the city of Irbid

The project aims to develop the city and make it a smart city, in addition to solving the problem of parking and traffic congestion, which leads to serving the city's population and provide job opportunities and achieve a profitable return to the entrepreneur

Based on the expected market share, the expected annual income is 393,120 JD in the first year, rising to 2,564,662 JD in the tenth year, and the expected pay pack period is 4 years with an internal rate of return of 34.20%.

the following table shows the main results of the study:

Table1 main results of the study

Results	
Total investment cost (JD)	2,448,446
IRR (%)	%34.20
Payback period (years)	4
Net present value (for 10 years)	7,336,366
Provided job opportunities	8
Expected total cash inflows (for 10 years)	14,938,938
Total operating expenses (for 10 years)	3,051,081
Total net profit (for 10 years)	7,740,983

2. Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

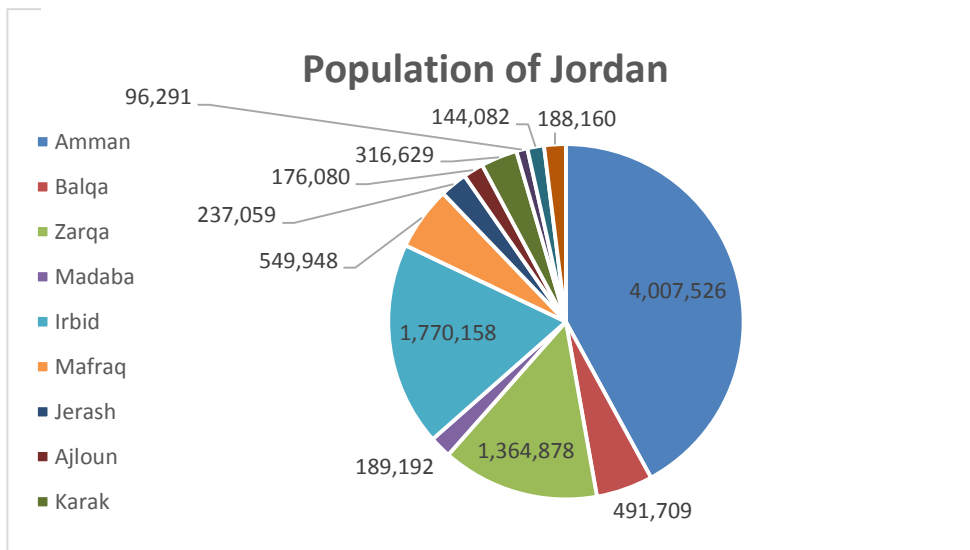
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafrq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table 2 Population Distribution according to Age Group

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

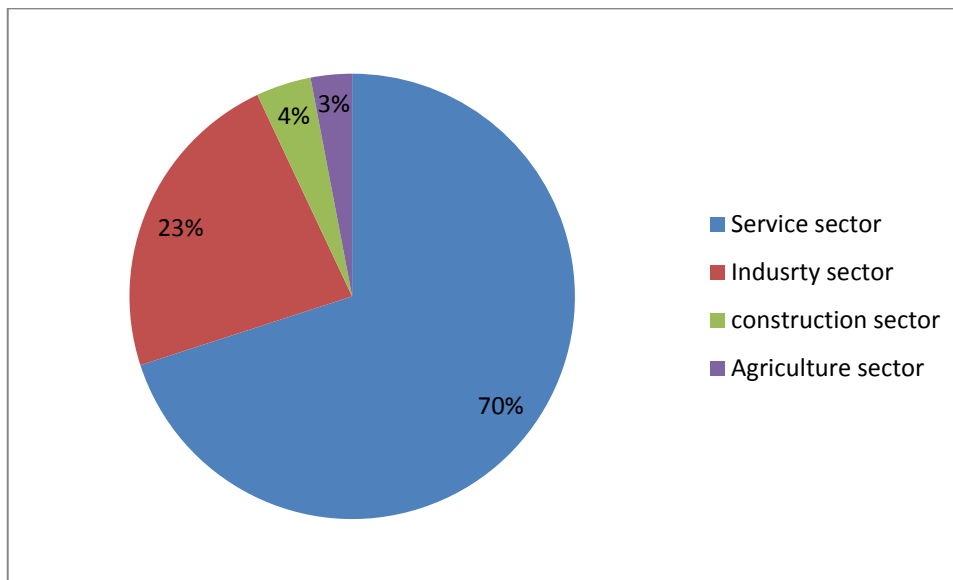
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure 2 Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and comprising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and

medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

furthermore, in its endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

3. Market Analysis

Project Description

The proposed project aims to create multi-storey automatic parking buildings or so-called smart parking in Irbid. where its spaces have a large number of advantages. The project was adopted to solve the problem of lack of parking due to the lack of commercial land available in the city of Irbid with a large number of vehicles. The project can reduce traffic jam in the city due to increasing numbers of vehicles every year.

It is suggested to build up the the automatic parking in the place that is crowded most of the time, due to the existence of companies, malls, banks, and restaurants. capacity of the project is estimated at 300 parking lot



Project objectives

The objectives of the project can be summarized in the following points:

- Exploitation of available lands in the city of Irbid to facilitate one of the life necessities in the region. Make it easier for drivers and save time looking for parkings and reducing noise.
- Upgrading the city towards an intelligent city.
- Reduces the traffic jam in Irbid city.
- Providing job opportunities.
- Achieving high profit from the project.

Proposed Service

The proposed project provides the service of automatic parking, the parking consists of multiple floors to floors to accommodate the largest number of vehicles. The price of the parking is considered to be competitor in the market, the project saves time and effort for vehicle owners.

So that the project consists of several floors to accommodate the largest number of vehicles, taking into account the provision of a suitable price for owners of vehicles. The project saves time and effort for vehicle owners

Target Segments

The project targets the residents of Irbid city in general, including workers in public and private sector, and visitors to commercial centers and restaurants.

Market size Analysis

The analysis of market size depends on the population of Irbid city, depending on the results of General Census of Population and Housing 2015.

Table 3 population of irbid city

Irbid city	City population 2015	Jordanians	Non-Jordanians
numbers	1,770,158	1,316,618	453,540

Source: department of statistics

Table 4 number of vehicles in irbid city

Property	Salon	Buses	shipping	tanks	Trucks	others	Total
Private	76,069	1,031	19,962	253	149	8,486	105,914
public	2,430	1,174	1,488	780	3,328	33	9,233

Source: department of statistics

1.1 Competitors analysis

The analysis of competitors is divided into two parts, direct competitors and indirect competitors n. There isn't an automatic parking in Irbid city, which means there isn't any direct competition. Traditional parkings are the indirect competitors for the project.

Project Capacity

The following table shows the expected capacity for the automatic parking:

Table 5 project capacity

Project capacity	
Parking capacity	300
Daily capacity	4,200
Annual capacity	1,512,000
Total capacity	1,516,500

Market Share

Due to the increase of vehicles and companies, malls, Institutions, the need for parking lots is increasing.

The following table shows the expected market share for 10 years:

Table 6 project expected market share during (2018-2027)

Year	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Percentage	%10	%15	%20	%25	%30	%35	%40	%45	%50	%50

Prices analysis and pricing policy

Based on studying and analyzing the prices of similar project, the suggested prices are as follow:

- Less than 2 hours: 2 JD/hour
- All day: 4 JD/day

The suggested prices are considered to be lower than similar parkings and competitive to the traditional parking.

Expected Income Analysis

The following table shows the expected revenues for 10 years (2018-2027)

Table 7 expected revenues for 10 years (2018-2027)

Year	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
1 hour price	2.00	2.06	2.12	2.19	2.25	2.32	2.39	2.46	2.53	2.61
All day price	4.00	4.12	4.24	4.37	4.50	4.64	4.78	4.92	5.07	5.22
Percentage of 1 hr	70%	70%	70%	70%	70%	70%	70%	70%	70%	70%
Percentage of all day	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%
Occupancy rate	10.00%	15.00%	20.00%	25.00%	30.00%	35.00%	40.00%	45.00%	50.00%	50.00%
Annual income	393120	607370	834122	1073932	1327380	1595068	1877623	2175696	2489963	2564662

4. Technical Analysis

Technical feasibility study aims to develop technical specifications for the project, depending on the results of the market study, in addition to the assessment of the most important investment and operational cost items.

The technical study of the project includes the explanation and description of the aspects of the project and its components in addition to the expense of construction and operational requirements for the project and determining the general specifications for the costs, according to the estimates by the technical team of the Center.

Required Human Resources

Table 8 required human resources

Job title	Number
Indirect employees	
General manager	1
Accountant	1
Cleaner	2
Total indirect employees	4
Direct employees	
Technical operator	2
Maintenance technician	1
Security	1
Total direct employees	4
Total	8

The following shows the job description:

Location Analysis

Irbid city is characterized by the highest population density in the Hashemite Kingdom of Jordan; It has the largest number of residents in terms of its area. The increase of the city population led to the increase of number of vehicles each year leads to traffic congestion, which requires finding a solution. The need for similar project in Irbid city is essential. The suggested location is Amman complex which is known as an active location, containing companies, institutions, banks, malls, restaurants, for that reasons the location considered to be strategic for an automatic parking.

Technical Requirments Analysis

1.1.1 land and construction works

the project needs a land with an area of 1000 m² in Amman complex in Irbid city, with an estimated value of 80,000 Jordanian Dinars.

Table9 required land for the project

Land	Area (m ²)
Needed land area	1,000

- The construction work is summarized in the following points: the cost of construction work vary based on the technologies that is used and country of manufacture and supply. The study included the collection of data about many systems from various developed countries and reached possibility of dividing the cost of construction (according to the systems and devices supplied) into three categories:
 1. Advanced systems and technology and the estimated cost of 26,000 JD per one parking.
 2. Advanced technology systems but less complex with an estimated cost of 11,000 JD per one parking.
 3. Medium technologies with a good specifications with an estimated cost of 7,000 JD per one parking.

- Any of the above mentioned catogries can be used based on the nature of the project, its needs, its owner and the allocated budger. For this proposed project, and after studying and analyzing, it's found that the third category suites the project objectives, which costs about 7,000 Jd per one parking.

The suggested capacity is 300 parking lot; therefore, the total estimated cost is 2,100,000 JD including all construction work and preparation of the land in addition to the controlling system and computers.

Machinery and equipment

The following table shows needed machines and equipment:

Table 10 required machinery and equipment

Machinery and equipments	Qty
Fire fighting equipments	2
Office air conditioner	2

Furniture

the project needs simple furniture such as office furniture and chairs.

5. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 12.10% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Income tax on the net project income is 20 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 11 General Assumptions

General Information	
All Financial Numbers (Currency)	Jordanian Dinar
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 12 Currency Exchange Rates

Exchange Rates	Jordan dinars
American Dollar	0.708
Euro	0.760

Table 13 working time

Working time	
No. of weeks	52
No. of Working days in the week	7
No of days	360
No. of Working hours	8

Table 14 Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	3.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 15 Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	10.00%
Utilities Expenses from Total Revenues	5.00%
Maintenance Expenses from Total Revenues	1.00%
Depreciation Expenses from Total Revenues	0.50%
Insurance Expenses from Total Assets Value	0.50%
Marketing Expenses from Total Revenues	0.50%

Table 16 Income Tax Assumptions

Income Tax Assumptions	Value
(1) Average Income Tax in Jordan	20%
(2) Income Tax Deduction	0%
Income Tax after Deduction	20%
(3) Compulsary Reserve Percentage	10%
Other Assumptions	Value
Annual Monthly Salaries have been calculated after multiplying monthly	14

Table 17 Risk Premiun

Risk Premiun	
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	9.48%
(3) Market Risk Premiun	10.12%
Income Tax Rate	5.00%
(4) Beta	1
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 18 Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premium	%10.12
Income Tax Rate	%20
Bets	1.00
Equity	1,224,223
Loans Value	1,224,223
Loans	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	%7.58
Loans Value	1,224,223
Loans Percentage	%50
Equity	
Cost of Equity	%16.6
Equity Value	1,224,223
Equity Percentage	%50
Gross	
Project Value	2,448,446
Weighted Average Cost of Capital	%11.23

Capital Expenditures

The estimated cost of the project is 2,448,446 JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 19 Land Capital Expenditure

Land	(Jordan Dinar/Meters Square) Cost	Area (M2)	Cost
Required Land	80	1,000	80,000
Cost			80,000

Table 20 Construction Work Capital Expenditure

Description	Cost (JOD per Squared	Area	Gross
Total cost	7000	300	2,100,000

Table 21 Machinery and equipment Capital Expenditure

Machinery and equipment	Number	Unit cost (JD)	Total cost (JD)
Fire fighting equipment	2	500	1,000
Air condition	2	400	800
Other	-	-	1,000
Total cost (JD)			2,800

Table 22 Furniture Capital Expenditure

Furniture	Number	Unit cost (JD)	Total cost (JD)
General furniture	-	-	2,000
Total			2,000

Table 23 Pre-Operating Expenses

Pre-operating expenses	Estimated cost (JD)
Governmental fees	1,000
Transportation	1,000
Marketing	500
Training	500
Total	3,000

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 24 Working Capital

Working Capital	No. of	%	Estimated Cost
Operating expenses	3	%25	24,878
General and administrative expenses	3	%25	3,731
Marketing expenses	3	%25	491
Indirect Salaries	3	%25	8,960
Total			38,060

Table 25 Expenses Summary

Capital expenditure	Cost (JD)	Contingency	Cost (JD)	%
Land	80,000	10%	88,000	%3.6
Construction works	2,100,000	10%	2,310,000	%94.3
Machinery & Equipment	2,800	10%	3,080	%0.1
Furniture	2,000	10%	2,200	%0.1
Vehicles	0	10%	0	%0.0
Pre-Operating Expenses	3,000	10%	3,300	%0.1
Working capital	38,060	10%	41,866	%1.7
Total (JD)	2,225,860		2,448,446	100%

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (2,448,446) JD, where 50% of the project will be financed through loans (1,224,223) JD, and the remaining (50%) through (Equity) with (1,224,223) JD. The following table summarizes the general structure of the required financing:

Table 26 Sources of Funding

Funding Sources	Amount	Percentage
Equity	1,224,223	%50
Loans	1,224,223	%50
Total	2,448,446	%100
Use of Fund	Amount	Percentage
Capital Expenditures	2,403,280	%98.16
Pre-Operating Expenses	3,300	%0.13
Working Capital	41,866	%1.71
Total	2,448,446	%100.00

Operating Expenses

The following table summarizes the expected operating expenses for the proposed project (2018-2027):

Table 27 Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials and Packaging	39,312	60,737	83,412	107,393	132,738	159,507	187,762	217,570	248,996	256,466
Utilities from Revenues	19,656	30,369	41,706	53,697	66,369	79,753	93,881	108,785	124,498	128,233
Maintenance from Revenues	3,931	6,074	8,341	10,739	13,274	15,951	18,776	21,757	24,900	25,647
Disposables from Revenues	1,966	3,037	4,171	5,370	6,637	7,975	9,388	10,878	12,450	12,823
Direct Salaries	25,600	26,650	27,742	28,880	30,064	31,296	32,579	33,915	35,306	36,753
Others	9,046	12,687	16,537	20,608	24,908	29,448	34,239	39,291	44,615	45,992
Total	99,511	139,552	181,910	226,686	273,990	323,931	376,626	432,196	490,765	505,915

General and Administrative Expenses

The following table summarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 28 General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	750	773	796	820	844	869	896	922	950	979
Hospitality	600	618	637	656	675	696	716	738	760	783
Stationary and Printing	100	103	106	109	113	116	119	123	127	130
Transportation	100	103	106	109	113	116	119	123	127	130
Insurance	12,016	12,377	12,748	13,131	13,525	13,930	14,348	14,779	15,222	15,679
Miscellaneous	1,357	1,397	1,439	1,482	1,527	1,573	1,620	1,668	1,719	1,770
Total	14,923	15,371	15,832	16,307	16,796	17,300	17,819	18,353	18,904	19,471

Marketing Expenses

The following table shows marketing expenses

Table 29 Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	1,966	3,037	4,171	5,370	6,637	7,975	9,388	10,878	12,450	12,823

Human Resources

Automatic parking in Irbid

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance, etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 30 expected numbers of employees

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect Employees										
General Manager	1	1	1	1	1	1	1	1	1	1
Accountant	1	1	1	1	1	1	1	1	1	1
Cleaner	2	2	2	2	2	2	2	2	2	2
Total indirect employees	4	4	4	4	4	4	4	4	4	4
direct Employees										
Technical operator	2	2	2	2	2	2	2	2	2	2
Maintenance technician	1	1	1	1	1	1	1	1	1	1
Guard	1	1	1	1	1	1	1	1	1	1
Total direct employees	4	4	4	4	4	4	4	4	4	4
total	8	8	8	8	8	8	8	8	8	8

Table 31 Expected monthly Salaries and Wages (2027-2018)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect Employees										

General Manager	1,300	1,353	1,409	1,467	1,527	1,589	1,654	1,722	1,793	1,866
Accountant	500	521	542	564	587	611	636	662	690	718
Cleaner	220	229	238	248	258	269	280	291	303	316
direct Employees										
Technical operator	400	416	433	451	470	489	509	530	552	574
Maintenance technician	400	416	433	451	470	489	509	530	552	574
Guard	400	416	433	451	470	489	509	530	552	574

Table 32 Annual Expected Salaries

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect Employees										
General Manager	20,800	21,653	22,541	23,465	24,427	25,428	26,471	27,556	28,686	29,862
Accountant	8,000	8,328	8,669	9,025	9,395	9,780	10,181	10,599	11,033	11,485
Cleaner	7,040	7,329	7,629	7,942	8,268	8,606	8,959	9,327	9,709	10,107
Total indirect salaries	35,840	37,309	38,839	40,432	42,089	43,815	45,611	47,481	49,428	51,455
direct Employees										
Technical operator	12,800	13,325	13,871	14,440	15,032	15,648	16,290	16,958	17,653	18,377
Maintenance technician	6,400	6,662	6,936	7,220	7,516	7,824	8,145	8,479	8,826	9,188
Guard	6,400	6,662	6,936	7,220	7,516	7,824	8,145	8,479	8,826	9,188
Total direct salaries	25,600	26,650	27,742	28,880	30,064	31,296	32,579	33,915	35,306	36,753
Total	61,440	63,959	66,581	69,311	72,153	75,111	78,191	81,397	84,734	88,208

Depreciations

The following tables illustrate cost of Capex and Depreciation rates:

Table 33 Capex and Depreciation Expenses

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	88,000	%0.0	%0.0
Construction Works	2,310,000	%5.0	%0.0
Machinaries	3,080	%10.0	%2.0
Furniture	2,200	%10.0	%5.0
Vehicles	0	%10.0	%0.0
	2,403,280		

Table 34 Depreciations and Additions on Construction Works

Capex, Annual Additions and depreciations	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	2,403,28	2,403,45	2,403,63	2,403,81	2,404,00	2,404,20	2,404,41	2,404,63	2,404,85	2,405,09
Total Depreciation Expenses	116,028	116,045	116,063	116,082	116,101	116,121	116,142	116,163	116,186	116,209
Total Accumulated Depreciation	116,028	232,073	348,136	464,218	580,318	696,439	812,581	928,744	1,044,93	1,161,14
Total Additions	0	172	178	185	193	200	208	217	226	235
Total Net Book Values	2,287,25	2,171,37	2,055,49	1,939,59	1,823,69	1,707,76	1,591,83	1,475,88	1,359,92	1,243,95

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table 35 Loan Details

Loan Value		1,224,223		
Annual Interest Rate		9.48%		
Loans Period		5		
Grace Period		0		
Loan Starts at year:		2018		
Annual Payment		318,668		
Number of Payments		5		
Year	Annual Payment	Interest	Capital	Loan Remaining Value
2018				1,224,223
2019	318,668	116,056	202,612	1,021,611
2020	318,668	96,849	221,819	799,792
2021	318,668	75,820	242,848	556,944
2022	318,668	52,798	265,870	291,074
2023	318,668	27,594	291,074	0

Income Statement

Table 36 Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	393,120	607,370	834,122	1,073,93	1,327,38	1,595,06	1,877,62	2,175,69	2,489,96	2,564,66
Gross Operating Revenues	393,120	607,370	834,122	1,073,93	1,327,38	1,595,06	1,877,62	2,175,69	2,489,96	2,564,66
Operating Expenses	(99,511)	139,552)	181,910)	226,686)	273,990)	323,931)	376,626)	432,196)	490,765)	505,915)
Gross Operating Profit	293,609	467,818	652,212	847,246	1,053,39	1,271,13	1,500,99	1,743,50	1,999,19	2,058,74
<i>Gross Profit Percentage</i>	<i>%75</i>	<i>%77</i>	<i>%78</i>	<i>%79</i>	<i>%79</i>	<i>%80</i>	<i>%80</i>	<i>%80</i>	<i>%80</i>	<i>%80</i>
Salaries and Benefits (Indirect Staff)	(35,840)	(37,309)	(38,839)	(40,432)	(42,089)	(43,815)	(45,611)	(47,481)	(49,428)	(51,455)
General and Administraive	(14,923)	(15,371)	(15,832)	(16,307)	(16,796)	(17,300)	(17,819)	(18,353)	(18,904)	(19,471)
Markeing Expenses	(1,966)	(3,037)	(4,171)	(5,370)	(6,637)	(7,975)	(9,388)	(10,878)	(12,450)	(12,823)
Pre Operating Expenses	(3,300)									
Gross Indirect Expenses	(56,029)	(55,717)	(58,842)	(62,108)	(65,522)	(69,090)	(72,818)	(76,713)	(80,782)	(83,749)
Income before Interest,	237,580	412,101	593,371	785,138	987,868	1,202,04	1,428,17	1,666,78	1,918,41	1,974,99
Fixed Assets Depreciations	116,028)	116,045)	116,063)	116,082)	116,101)	116,121)	116,142)	116,163)	116,186)	116,209)
Income before Tax and Interests	121,552	296,056	477,308	669,056	871,768	1,085,92	1,312,03	1,550,62	1,802,23	1,858,78
Bank Interests	116,056)	(96,849)	(75,820)	(52,798)	(27,594)	(0)	(0)	(0)	(0)	(0)
Income Before Tax	5,496	199,207	401,488	616,258	844,174	1,085,92	1,312,03	1,550,62	1,802,23	1,858,78
Income Tax	(1,099)	(39,841)	(80,298)	123,252)	168,835)	217,185)	262,407)	310,125)	360,446)	371,758)

Net Profit	4,397	159,366	321,190	493,006	675,339	868,741	1,049,63	1,240,49	1,441,78	1,487,03
<i>Net Profit Percentage</i>	<i>%1</i>	<i>%26</i>	<i>%39</i>	<i>%46</i>	<i>%51</i>	<i>%54</i>	<i>%56</i>	<i>%57</i>	<i>%58</i>	<i>%58</i>
Compulsory Reserves	(440)	(15,937)	(32,119)	(49,301)	(67,534)	(86,874)	104,963)	124,050)	144,178)	148,703)
Retained Earnings	3,957	147,386	436,457	880,163	1,487,96	2,269,83	3,214,50	4,330,95	5,628,55	6,966,88

Expected Cashflows Statement

The table below summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 37 Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash Inflows from Operating										
Net Profit	4,397	159,366	321,190	493,006	675,339	868,741	1,049,63	1,240,49	1,441,78	1,487,03
Bank Interests	116,056	96,849	75,820	52,798	27,594	0	0	0	0	0
Depreciation	116,028	116,045	116,063	116,082	116,101	116,121	116,142	116,163	116,186	116,209
Total Operating Cashflows before	236,481	372,260	513,073	661,886	819,034	984,862	1,165,77	1,356,66	1,557,97	1,603,24
Inventory (Increase/Decrease)	(3,276)	(1,785)	(1,890)	(1,998)	(2,112)	(2,231)	(2,355)	(2,484)	(2,619)	(622)
Accounts Payables	0	0	0	0	0	0	0	0	0	0
Working Capital	10,033	3,103	3,280	3,466	3,660	3,862	4,073	4,294	4,524	1,280
Net Cashflows from Operating	6,757	1,317	1,391	1,468	1,548	1,631	1,719	1,810	1,905	658

Cashflows from Investments	243,238	373,577	514,464	663,354	820,581	986,493	1,167,49	1,358,47	1,559,87	1,603,89
Fixed Assets (Procurement)										
Net Cashflows from Investments	)	(172)	(178)	(185)	(193)	(200)	(208)	(217)	(226)	(235)
Cashflows from Financing	)	(172)	(178)	(185)	(193)	(200)	(208)	(217)	(226)	(235)
Capital	1,224,223									
Loan Amortization	(202,612)	)	)	)	)	0	0	0	0	0
Bank Interest Rate	(116,056)	(96,849)	(75,820)	(52,798)	(27,594)	(0)	(0)	(0)	(0)	(0)
Loans	1,224,223	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing	2,129,778	)	)	)	)	0	0	0	0	0
Net (Increase/Decrease) in Cash	(30,264)	54,737	195,618	344,500	501,720	986,293	1,167,28	1,358,25	1,559,65	1,603,66
Cashflows at the Beginning of	0	(30,264)	24,473	220,091	564,591	1,066,31	2,052,60	3,219,88	4,578,14	6,137,79
Cashflows at the End of Period	(30,264)	24,473	220,091	564,591	1,066,31	2,052,60	3,219,88	4,578,14	6,137,79	7,741,45

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 38 Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current Assets										
Cash	(30,264)	24,473	220,091	564,591	1,066,312	2,052,605	3,219,886	4,578,142	6,137,792	7,741,455
Inventory	3,276	5,061	6,951	8,949	11,062	13,292	15,647	18,131	20,750	21,372
Total current Assets	(26,988)	29,535	227,042	573,541	1,077,373	2,065,897	3,235,533	4,596,273	6,158,541	7,762,827
Non Current Assets										
Fixed Assets (net)	2,287,252	2,171,378	2,055,494	1,939,598	1,823,690	1,707,769	1,591,836	1,475,889	1,359,929	1,243,954
Total Non Current	2,287,252	2,171,378	2,055,494	1,939,598	1,823,690	1,707,769	1,591,836	1,475,889	1,359,929	1,243,954
Total Assets	2,260,264	2,200,913	2,282,536	2,513,138	2,901,063	3,773,666	4,827,369	6,072,162	7,518,470	9,006,781
Current Liabilities										
Paybles	10,033	13,136	16,416	19,882	23,542	27,404	31,477	35,771	40,295	41,575
Remaining amount of	221,819	242,848	265,870	291,074	(0)	(0)	(0)	(0)	(0)	0
Total current	231,853	255,984	282,286	310,956	23,542	27,404	31,477	35,771	40,295	41,575
Non Current										
Long Terms Loans	799,792	556,944	291,074	0	0	0	0	0	0	0
Total Long Term	799,792	556,944	291,074	0	0	0	0	0	0	0
Total Liabilities	1,031,644	812,928	573,360	310,956	23,542	27,404	31,477	35,771	40,295	41,575
Owners Equity										

Shareholders	1,224,223	1,224,223	1,224,223	1,224,223	1,224,223	1,224,223	1,224,223	1,224,223	1,224,223	1,224,223	1,224,223
Statutory Reseve	440	16,376	48,495	97,796	165,330	252,204	357,167	481,217	625,395	774,098	
Retained Profits	3,957	147,386	436,457	880,163	1,487,968	2,269,835	3,214,502	4,330,951	5,628,557	6,966,885	
Total Equity	1,228,620	1,387,985	1,709,175	2,202,182	2,877,521	3,746,262	4,795,892	6,036,391	7,478,175	8,965,206	
Total Liabilities and	2,260,264	2,200,913	2,282,536	2,513,138	2,901,063	3,773,666	4,827,369	6,072,162	7,518,470	9,006,781	

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans; therefore, the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 39 Free Net Cashflows Table

Net Cashflows	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final value
Net Free Cashflows	(2,448,446)	246,5	373,4	514,2	663,1	820,3	986,2	1,167,2	1,358,2	1,559,6	1,603,6	20,585,1
Discount Factor	1.00	0.80	0.71	0.63	0.56	0.50	0.45	0.40	0.36	0.32	0.28	0.28
Net Present Value for	(2,448,446)	196,1	265,0	325,6	374,5	413,3	443,3	468,02	485,80	497,61	456,42	5,858,78

The table below illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates a total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 40 Payback Period

Payback Period	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free	(2,448,446)	246,538	373,405	514,286	663,1	820,38	986,293	1,167,28	1,358,25	1,559,65	1,603,66
Project Value	2,448,446	2,201,9	1,828,5	1,314,2	651,0	)	)	)	)	)	)
Payback Period	4	1	1	1	1	0	0	0	0	0	0

Table 41 Financial Analysis Results

Weighted Average Cost of Capital (WACC)	%12.10
Net Present Value for Cashflows	7,336,366
Payback Period	4
Internal Rate of Return	%34.20

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Table 42 Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%34.20	4	12.10%
Revenues declined by 10%	%32.10	5	12.10%
Operating Expenses Increased by 10%	%32.67	4	12.10%

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 7,336,366 Jordan Dinars considering that the project provides 8 Job Opportunities for the governorate residents.